

AI Practitioner – Amazon Bedrock

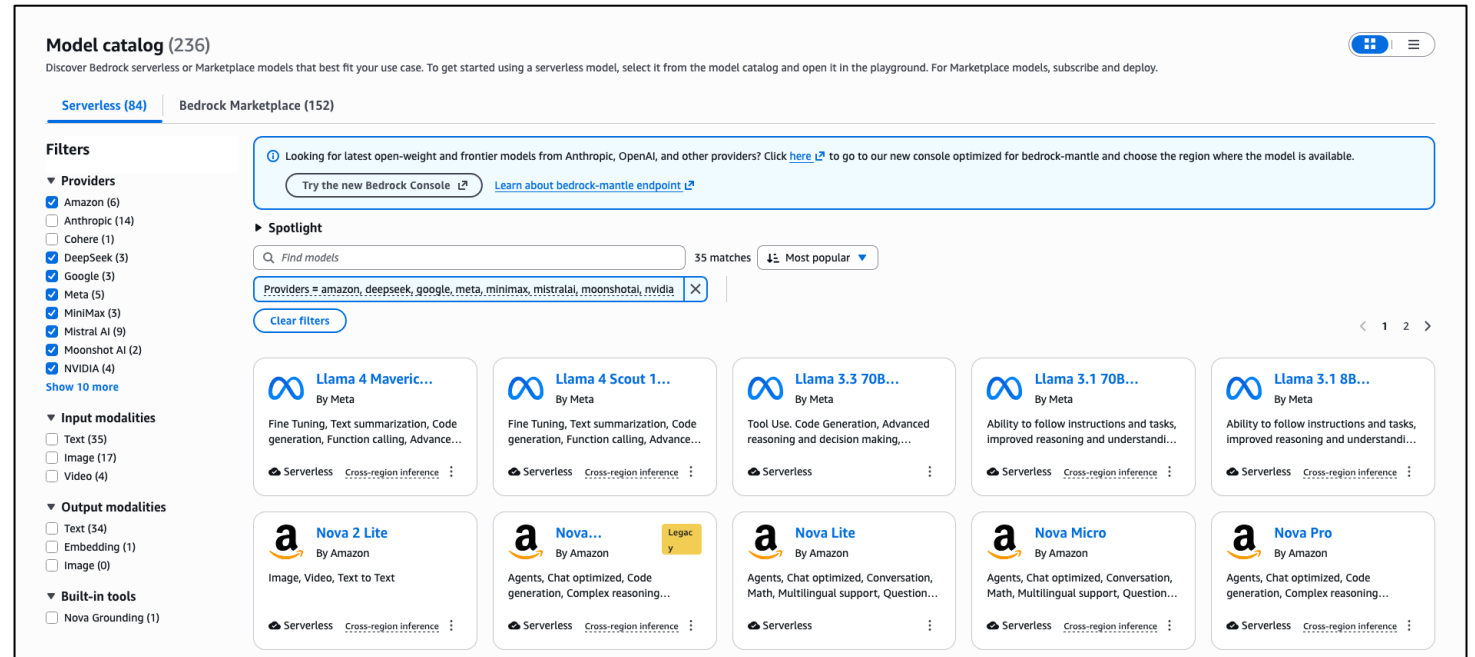
- Move from theory into practical Bedrock usage
- Choose a foundation model
- Send prompts and improve them with prompt engineering
- Configure inference parameters
- Use the Bedrock API
- Introduce fine-tuning and a basic application architecture



Amazon Bedrock

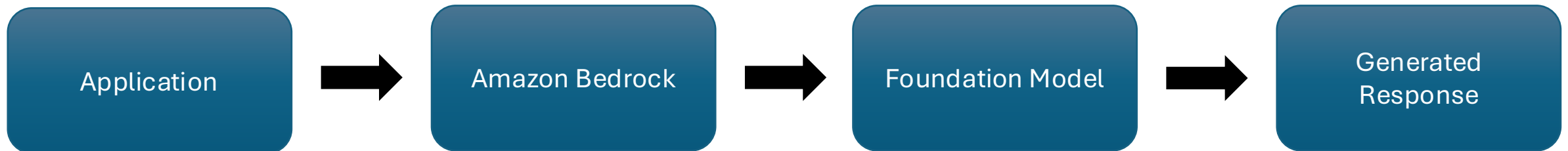
Getting Started With Amazon Bedrock

- Fully managed AWS service
- Access foundation models through APIs
- Compare capability, cost, context window, and performance
- Simple text generation may not need the largest model
- Multimodal applications may need a multimodal model



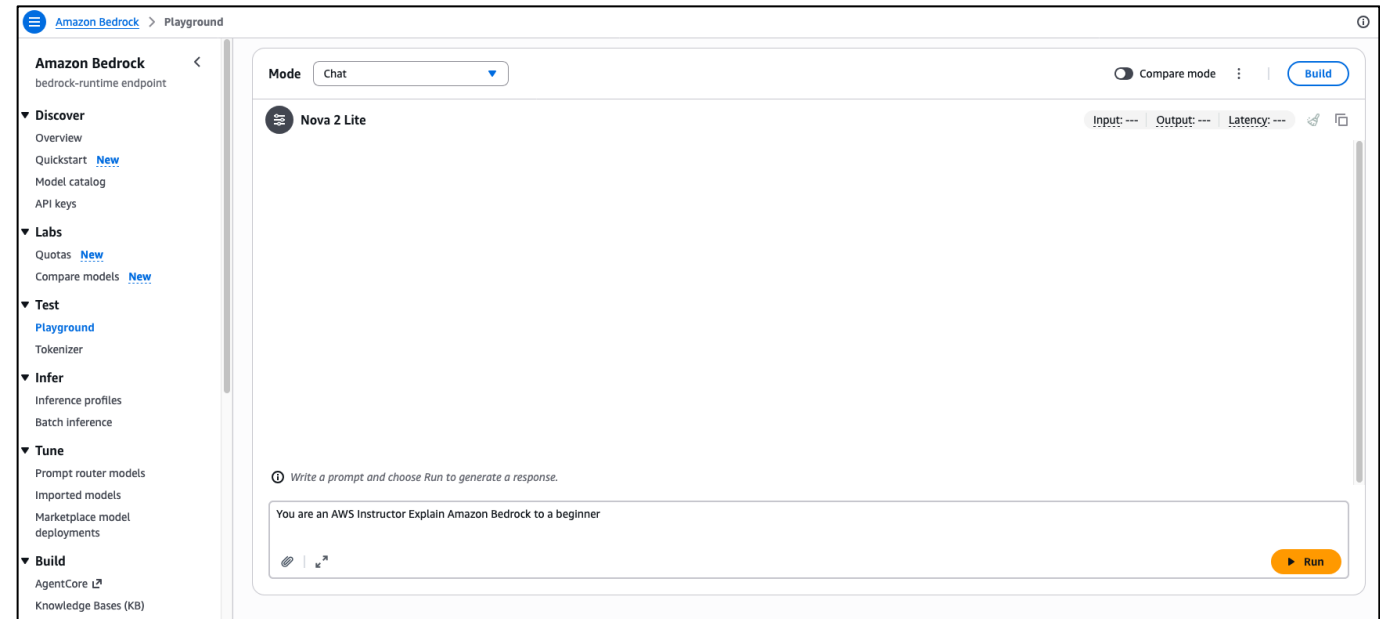
Sending a Prompt

- Example: “What is Amazon Bedrock?”
- The application sends the request to Bedrock.
- Bedrock sends it to the selected foundation model.
- The response returns to the application.



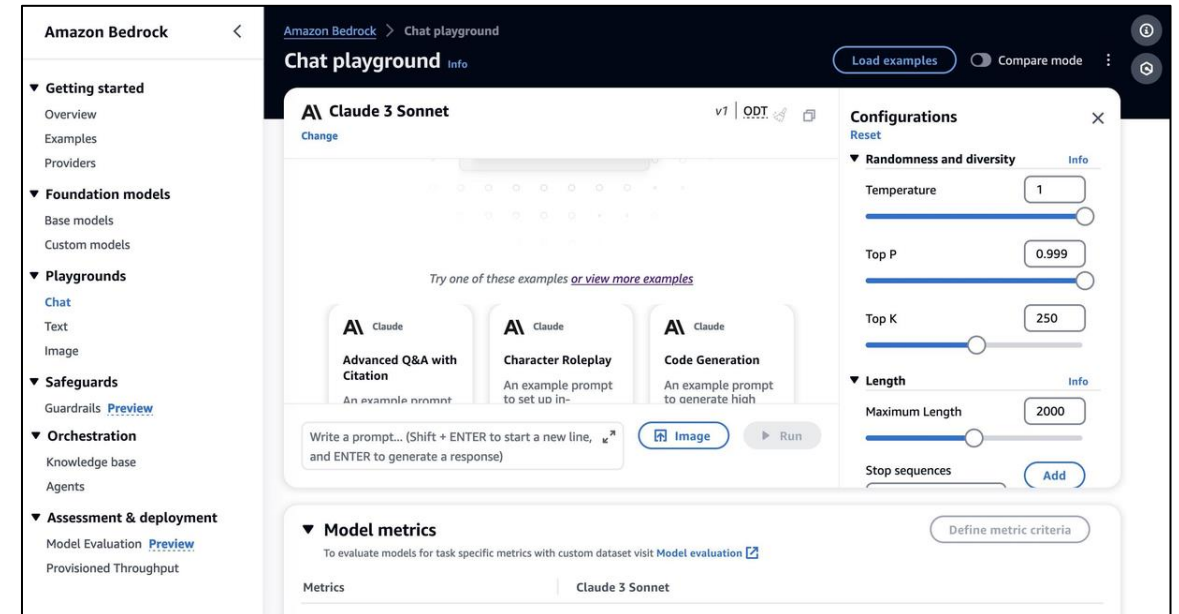
Prompt Engineering

- Weak: “Tell me about AWS.”
- Stronger: “You are an AWS instructor. Explain Amazon Bedrock to a beginner.”
- Request a definition, three benefits, and one real-world example.
- Clear instructions can improve response quality.



Configuring Inference Parameters

- Temperature: lower = generally more predictable; higher = more variation
- Maximum output tokens: controls response length
- Top-P and Top-K: influence token selection
- Not every model supports the same parameters



Using the Amazon Bedrock API

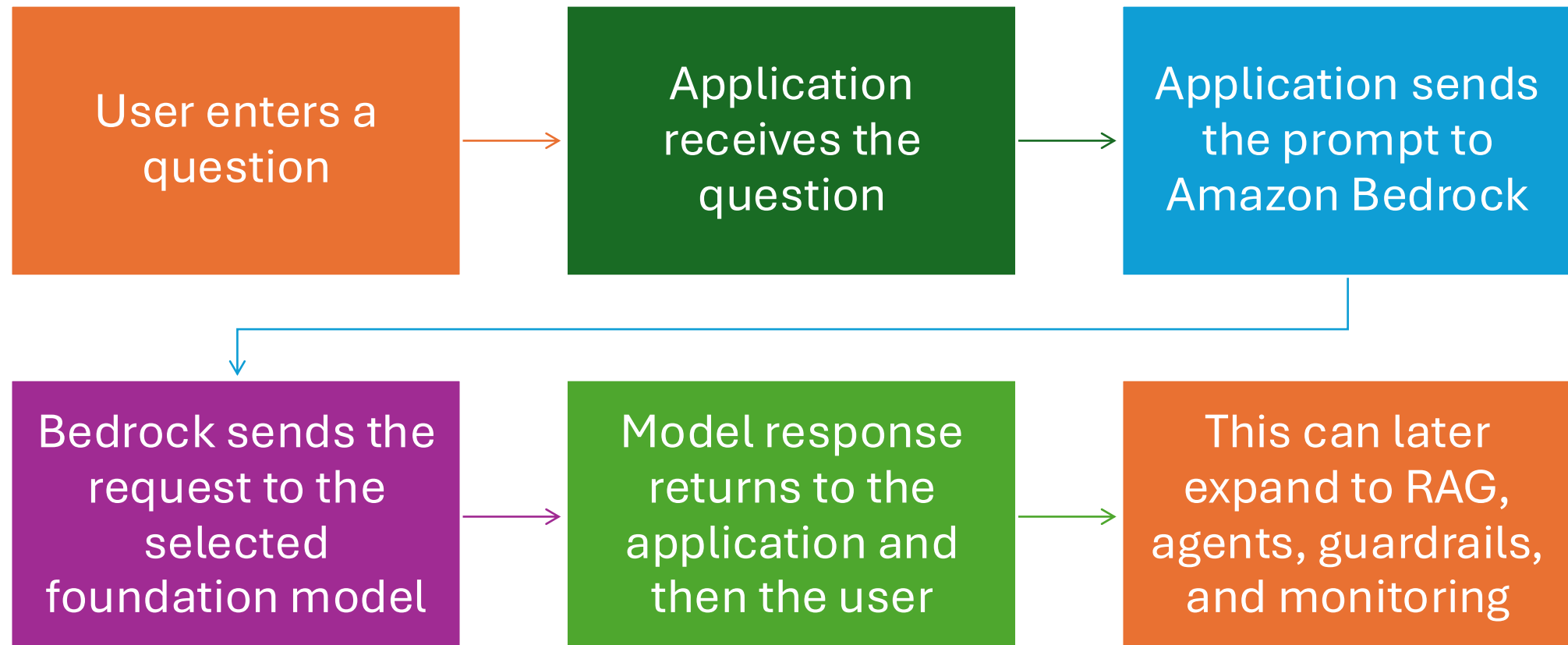
```
inferenceConfig={  
  "maxTokens": 500,  
  "temperature": 0.5,  
  "topP": 0.9  
}
```

- modelId
- prompt / messages
- inferenceConfig
 - └─ maxTokens
 - └─ temperature
 - └─ topP

Fine-Tuning

- Adapt a copy of a foundation model
- Model weights are updated
- Training data can contain labeled examples
- Example: customer questions paired with ideal responses
- Not every foundation model supports fine-tuning

Building the Basic Application



Video 2 Recap

- Choose a foundation model based on the use case
- Send prompts through Amazon Bedrock
- Use prompt engineering to provide clear instructions and context
- Configure supported inference parameters
- Use the Bedrock API to integrate models into software
- Understand when fine-tuning is appropriate — and when RAG may be better